

Mingxin Ding, PhD

Postdoctoral Associate · Department of Psychology & Neuroscience Program · Brandeis University

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EDUCATION

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| PhD, Neuroscience | Aug 2018 – Jan 2025 |
| Brandeis University, Waltham, MA · Advisor: Shantanu P. Jadhav | |
| MS, Biology | Aug 2015 – Jan 2017 |
| New York University, New York, NY · Advisors: Brendon O. Watson & György Buzsáki | |
| BS, Biological Sciences | Sep 2011 – Jun 2015 |
| Peking University, Beijing, China | |

RESEARCH EXPERIENCE

Postdoctoral Associate

Jan 2025 – Present

Brandeis University, Waltham, MA · *Jadhav Lab, Department of Psychology*

- Develop LFP and spike recording and analysis pipelines to quantify sleep architecture and oscillatory dynamics (theta, spindle, ripple events) and study memory-related thalamocortical and cortico-hippocampal connectivity in rodent models of neurodevelopmental disorders.
- Design and execute in vivo neurophysiology experiments in freely behaving rodents, integrating multi-tetrode recordings, fiber photometry, video/behavioral tracking, and EMG to study neuromodulatory mechanisms during sleep and learning.
- Perform stereotactic surgeries for chronic electrode and optical fiber implants, and targeted AAV/Cre-dependent viral injections for monitoring defined neural populations.
- Build end-to-end multimodal data pipelines (LFP/EEG, EMG, photometry, video) including QC, artifact handling, spectral feature extraction, statistical modeling, and cross-session validation.
- Apply reinforcement learning algorithms to multi-region neural data to model reward-based learning and adaptive behavior.
- Collaborate with human clinical research groups to align rodent physiology findings with questions in sleep, arousal, memory, and neurodevelopmental disorders.

Graduate Researcher

Aug 2018 – Jan 2025

Brandeis University, Waltham, MA · *Jadhav Lab, Department of Psychology*

- Led full experimental lifecycle of chronic, multi-region in vivo electrophysiology studies in freely moving rats: behavioral task design, stereotactic implantation, chronic recording, optogenetic tagging, data analysis, and manuscript preparation.
- Developed and optimized simultaneous single-unit and LFP recordings from hippocampal CA1, medial prefrontal cortex, and ventral tegmental area to study circuit mechanisms of learning, reward processing, and cognitive flexibility.
- Applied optogenetic tagging to identify VTA dopamine neurons and characterize cell-type-specific activity during trial-by-trial reward prediction and rule switching.
- Built signal-processing pipelines for high-frequency neural data: filtering, spike sorting (MountainSort4 + curation), occupancy-normalized firing maps, LFP spectral analysis (PSD/STFT/wavelets), spike-field coupling, and statistical validation.

- Applied machine learning and statistical methods — contrastive embeddings, kNN/linear decoders, GLM/point-process models, Bayesian decoding, mixed-effects models, permutation testing — to decode latent cognitive states and quantify population dynamics during uncertainty and behavioral adaptation.
- Designed scalable, reproducible analysis pipelines for multi-region, multi-session, multi-animal datasets with held-out validation and standardized performance benchmarking.
- Published first-author work in *Journal of Neuroscience* (2025, cover article) demonstrating coordinated dopamine, hippocampal, and prefrontal dynamics during rule updating and flexible behavior.

Lab Specialist

Jul 2017 – Jul 2018

University of Michigan, Ann Arbor, MI · *Watson Lab, Department of Psychiatry*

- Developed software suite for automated texture-discrimination behavioral paradigms integrated with real-time acute silicon probe recordings to study somatosensory coding and arousal-related dynamics in head-fixed mice.
- Performed surgeries, head-fixed electrophysiology experiments, and data-acquisition troubleshooting for in vivo neurophysiology studies.
- Built data cleaning, QC, and analysis workflows for behavioral and neural recordings, supporting analysis of motoric, arousal, and neurophysiological metrics during sensory-guided behavior.

Research Associate

Jan 2016 – Jun 2017

New York University, New York, NY · *Watson Lab & Buzsáki Lab, Neuroscience Institute*

- Conducted rodent behavioral experiments assessing spatial memory, motor function, and depression-related phenotypes.
- Developed statistical workflows to characterize spike–EEG/LFP coupling across cortical regions during awake and sleep states.
- Validated sleep-stage classification pipelines using multimodal physiological features (movement, LFP/EEG, EMG), benchmarking algorithmic outputs against expert manual scoring.

PUBLICATIONS

1. **Ding, M.**, Tomsick, P. L., Young, R. A., & Jadhav, S. P. (2025). Ventral tegmental area dopamine neural activity switches simultaneously with rule representations in the medial prefrontal cortex and hippocampus. *Journal of Neuroscience*, 45(37), e1670242025. ★ COVER ARTICLE
SPECIAL ISSUE: COMPUTATIONAL PROPERTIES OF PFC
2. Bueno-Junior, L. Z., Ghimire, A., **Ding, M.**, & Watson, B. O. (2025). Learning and arousal dynamics from motoric and neurophysiological metrics during a somatosensory task in mice. *eNeuro*.
3. Watson, B. O., **Ding, M.**, & Buzsáki, G. (2018). Temporal coupling of field potentials and action potentials in the neocortex. *European Journal of Neuroscience*, 48(7), 2482–2497.

PRESENTATIONS

1. **Ding, M.**, & Jadhav, S. P. (2023). Coordination of the hippocampus, prefrontal cortex and ventral tegmental area for spatial learning and flexible behavior. *Society for Neuroscience Annual Meeting*, Washington, DC.
2. **Ding, M.**, & Jadhav, S. P. (2019). Functions of hippocampal theta-entrained mPFC spiking activity. *Brandeis Life Science Volen Retreat*. Poster Award
3. **Ding, M.**, Watson, B. O., & Buzsáki, G. (2017). Temporal coupling of field potentials and action potentials in the neocortex. *Society for Neuroscience Annual Meeting*, Washington, DC.

TECHNICAL SKILLS

In vivo Neurophysiology Freely moving rodent electrophysiology; multi-site single-unit & LFP recordings (tetrode, silicon probe); fiber photometry; optogenetic tagging; multimodal signal synchronization

Surgery & Models

	Stereotactic surgery; chronic electrode & optical fiber implants; AAV/Cre-dependent viral strategies; peri-operative monitoring; rodent models of neurodevelopmental & neurological disorders
Behavioral Assays	Spatial working memory tasks (W maze, T maze); behavioral flexibility (rule switching); learning & memory assays; texture-discrimination; depression-related assays; sleep quantification (EEG/EMG/movement-based)
Signal Processing	Bandpass filtering; artifact handling; PSD/STFT/wavelet analysis; spike sorting (MountainSort4); event detection; time-frequency analysis; spike-field coupling; sleep-state classification
ML & Statistics	<i>Dimensionality reduction:</i> CEBRA, manifold learning, representation learning · <i>Decoding:</i> Bayesian decoding, kNN & linear decoders, GLM/point-process models · <i>Statistical inference:</i> mixed-effects models, permutation testing · <i>Sequence modeling:</i> HMM, state-space models · <i>Learning:</i> reinforcement learning
Programming	Python (NumPy, SciPy, Pandas, scikit-learn, PyTorch), MATLAB, C++ (familiar), R, SQL, Git/GitHub, AWS